Dielectric Materials, Polarization, Optical Properties

MO-201 Materials Science (Odd Sem 2026-27)

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Reference Textbooks

Materials Science & Engineering by V. Raghavan

Fundamentals of Materials Science & Engineering by W.D. Callister

Dielectric Materials & Polarization

- The large magnitude of the energy gap in an ideal insulator precludes the possibility of electrons being excited from the valence band to the conduction band by thermal means, much less so by an externally applied electric field

- Insulators are therefore very poor conductors of electricity and are known as **dielectrics**

- Dielectric materials find extensive use in the electrical industry for insulation purposes and as **capacitors**

- The charge stored in a capacitor increases if a dielectric material is inserted between the capacitor plates; this is due to the polarization in the dielectric

- There are various polarization mechanisms such as **electronic, ionic, orientation and space charge polarization**

- When an electric field is applied to a solid containing positive and negative charges, the positive charges are displaced in the direction of the field towards the negative end, while the negative charges are displaced in the opposite direction

- This displacement produces local dipoles throughout the solid

Dielectric Materials & Polarization

- The dipole moment per unit volume of the solid is the sum of all the individual dipole moments within that volume and is called the **polarization P** of the solid

- When a voltage V is applied to the capacitor, with the plates separated by vacuum, a charge develops on the plates; the capacitance C of the capacitor is a measure of this charge and is defined by

$$C = \frac{\varepsilon_0 A}{d}$$

  where ε_0 is the dielectric constant or permittivity of free space (vacuum), A is the area of the parallel plates and d is the distance of separation between them

- If a dielectric material is inserted between the plates, the charge on the plates increases due to polarization in the material and the capacitance is now given by

$$C = \varepsilon_r \varepsilon_0 \frac{A}{d}$$

  where ε_r is the relative dielectric constant or relative permittivity of the material

- ε_r and hence P can be determined by measuring the capacitance with and without the dielectric

Polarization Mechanisms

- **Electronic polarization** (Fig. 1a) is the result of the displacement of the positively charged nucleus and the (negative) electrons of an atom in opposite directions on application of an electric field

- When an electric field is applied to an ionic solid, cations and anions get displaced in opposite directions; the **ionic polarizability** is due to this shift of the ions relative to other oppositely-charged neighbours

- It should be distinguished from electronic polarization, where the electron cloud of an atom shifts with reference to its own nucleus

- Both are independent of T; however, ionic polarization is slower than electronic polarization because displacement of much heavier ion takes place as compared to electron cloud in latter

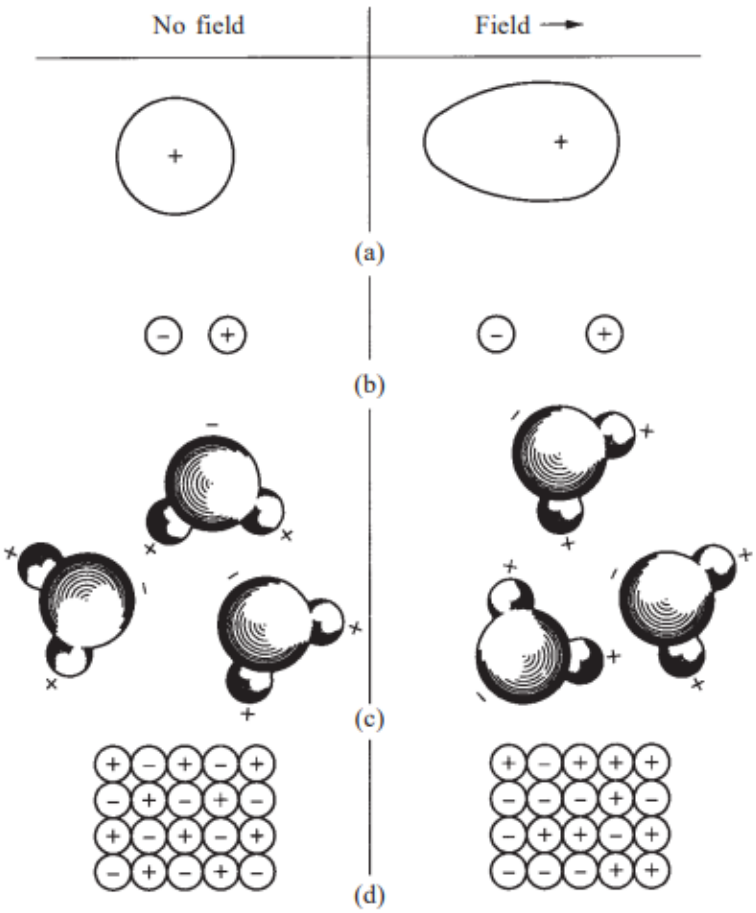


Fig. 1 Various polarization processes: (a) electronic (b) ionic (c) orientation and (d) space charge polarization

Polarization Mechanisms

- In CH_4 molecule, the centre of the -ve and +ve charges coincide, so that there is no permanent dipole moment; on the other hand, in CH_3Cl, the +ve and –ve charges do not coincide
- Since electronegativity of Cl > H, the Cl atom pulls the bonding electrons to itself more strongly than H; so, this molecule carries a dipole moment even without an electric field
- If electric field is applied on such molecules, they tend to align themselves in the applied field (just like atoms with permanent magnetic moments tend to align themselves with the applied magnetic field causing paramagnetism)
- The polarization due to this alignment is called **orientation polarization** and is T dependent
- With increasing T, the thermal energy tends to randomize the alignment

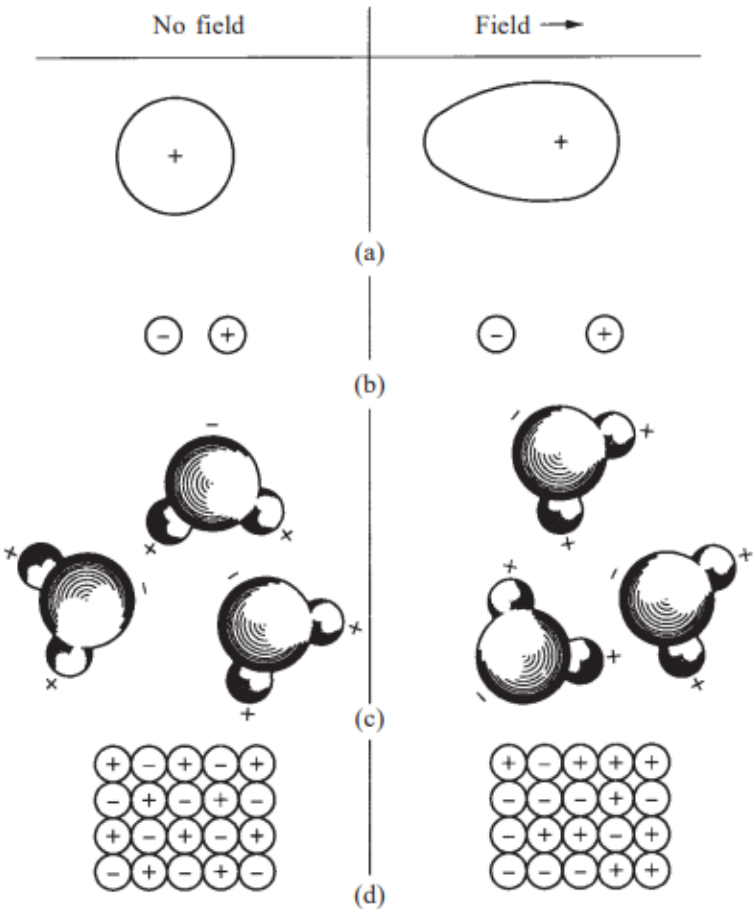


Fig. 1 Various polarization processes: (a) electronic (b) ionic (c) orientation and (d) space charge polarization

Polarization Mechanisms

- **Space charge polarization** occurs due to the accumulation of charges at the electrodes or at the interfaces in a multiphase material

- The ions diffuse over appreciable distances in response to the applied field, giving rise to a redistribution of charges in the dielectric medium

- The total polarization of a material is the sum of the contributions from the various sources

$$P_{\text{total}} = P_e + P_i + P_o + P_s$$

- Orientation polarization is even slower than ionic polarization

- Space charge polarization is the slowest process, as it involves the diffusion of ions over several interatomic distances

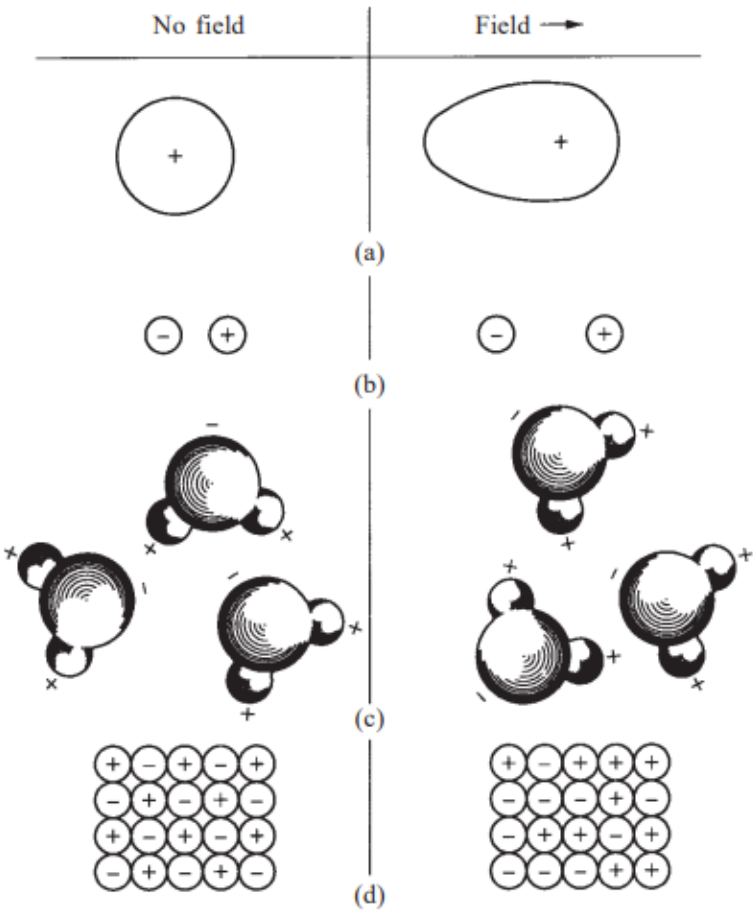


Fig. 1 Various polarization processes: (a) electronic (b) ionic (c) orientation and (d) space charge polarization

Ferroelectric Materials

- In **ferroelectrics**, the dielectric constants are some two orders of magnitude larger than those in ordinary dielectrics; Barium titanate is a ferroelectric with a relative dielectric constant of over 2000, compared to less than 10 for ordinary dielectrics

- The difference in the magnetic susceptibility between ferromagnetic and paramagnetic materials bears a direct analogy to this difference in the values of the dielectric constants

- As in the ferromagnetic phenomenon, the electric dipoles in a ferroelectric solid are all aligned in the same direction, even in the absence of an electric field

- Ferroelectric crystals always exhibit the **piezoelectric** property, which is the mechanical response of a crystal to an electric field or the electrical response to a mechanical stress

- The most important use of $BaTiO_3$ is as a piezoelectric crystal in applications such as microphones, phonograph pickups, strain gauges and sonar devices; the high dielectric constant of ferroelectric crystals is also useful for storing energy in small-sized capacitors in electrical circuits

Ferroelectric Materials

- Above 120°C, barium titanate is a cubic crystal in which the centres of the -ve and +ve charges coincide and there is no spontaneous dipole moment

- If the crystal is cooled to below the (ferroelectric) Curie temperature of 120°C, the Ti ion shifts to one side of the body centre and there is also a displacement of the neighbouring oxygen anions

- The crystal transforms from a cubic to a tetragonal structure on cooling through 120°C; the c/a ratio of the tetragonal cell is 4.03 Å/3.98 Å = 1.012

- The centres of the +ve and -ve charges do not coincide any longer and local dipoles are created throughout the crystal; the aligned dipoles of neighbouring unit cells result in a large polarization in the solid

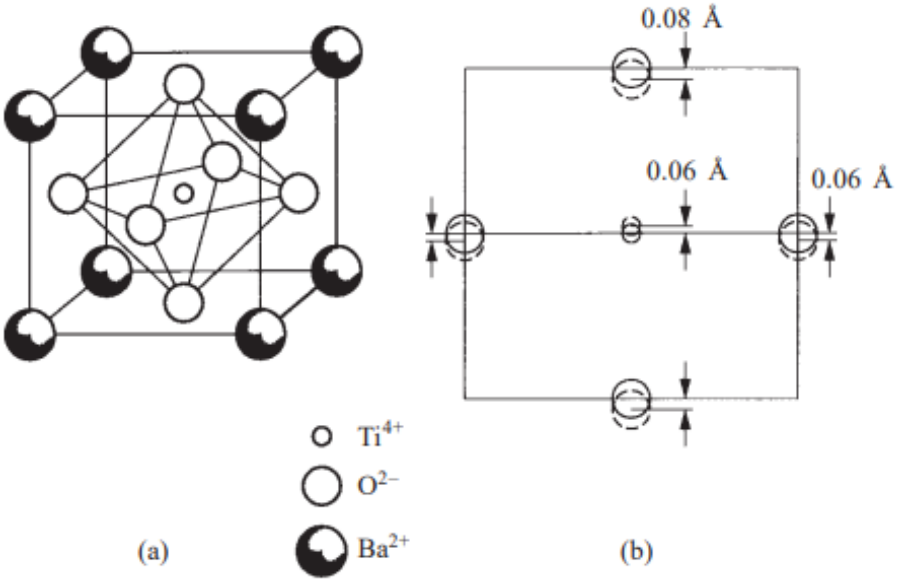


Fig. 2 (a) Cubic unit cell of $BaTiO_3$ crystal (b) the dashed circle in the middle of the front view shows the shifting of the Ti ion, on cooling through the Curie temperature; the shift of the oxygen anions is also shown

Optical Properties of Metals

- In both cases of metals shown in Fig. 3 (a & b), a high-energy band is only partially filled with electrons; metals are opaque because the incident radiation having frequencies within the visible range excites electrons into unoccupied energy states above the Fermi energy (Fig. 4a); as a consequence, the incident radiation is **absorbed**

- Total absorption is within a very thin outer layer, usually less than 0.1 μm; thus only metallic films thinner than 0.1 μm are capable of transmitting visible light; all frequencies of visible light are absorbed by metals because of the continuously available empty electron states, which permit electron transitions (Fig. 4a)

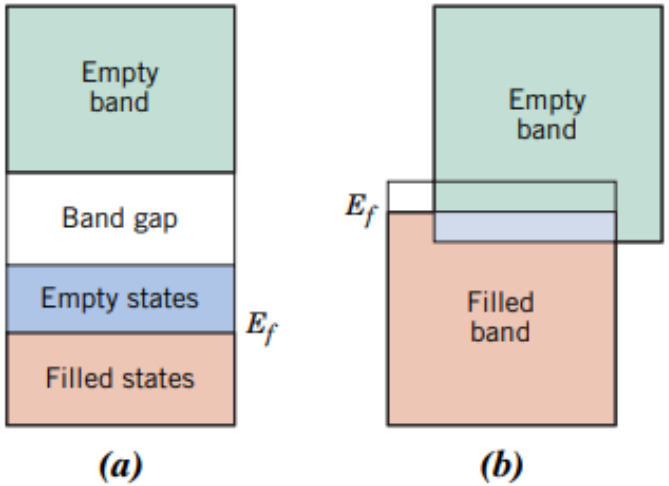


Fig. 3 Electron band structure of metals such as (a) Cu (b) Mg

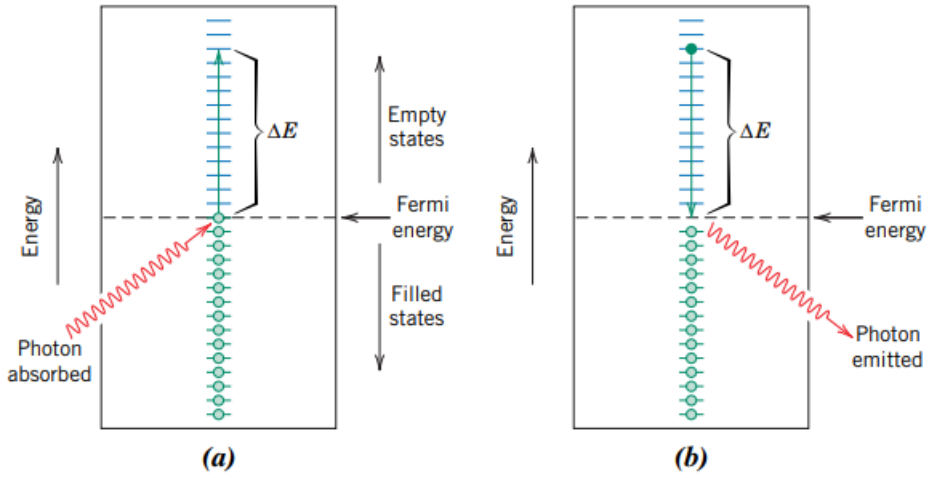


Fig. 4 (a) Mechanism of photon absorption (b) Reemission of a photon of light

Optical Properties of Metals

- In fact, metals are opaque to all electromagnetic radiation on the low end of the frequency spectrum from radio waves, through infrared and the visible, and into about the middle of the ultraviolet radiation; metals are transparent to high-frequency (x- and γ-ray) radiation

- Most of the absorbed radiation is reemitted from the surface in the form of visible light of the same wavelength, which appears as **reflected light**; an electron transition accompanying reradiation is shown in Fig. 4b; reflectivity for most metals is between 0.90 and 0.95; some small fraction of the energy from electron decay processes is dissipated as heat

- Because metals are opaque and highly reflective, the perceived color is determined by the wavelength distribution of the radiation that is reflected and not absorbed

- A bright silvery appearance when exposed to white light indicates that the metal is highly reflective over the entire range of the visible spectrum (i.e. frequency & number of reemitted photons in reflected beam approximately same as incident beam, e.g. Al and Ag)

- Cu and Au appear red-orange & yellow, respectively, because some of the energy associated with light photons having short λ is not reemitted as visible light

Optical Properties of Nonmetals

- By virtue of their electron energy band structures, nonmetallic materials may be transparent to visible light; therefore, in addition to **reflection** and **absorption**, **refraction** and **transmission** phenomena must also be considered

- Light that is transmitted into the interior of transparent materials experiences a decrease in velocity and, as a result, is bent at the interface; this phenomenon is termed **refraction**

- The phenomenon of refraction is related to electronic polarization at the relatively high frequencies for visible light

- Because the retardation of electromagnetic radiation in a medium results from electronic polarization, the size of the constituent atoms or ions has considerable influence on the magnitude of this effect — generally, the larger an atom or ion, the greater the electronic polarization, the slower the velocity, and the greater the index of refraction

- The index of refraction for a typical soda–lime glass is approximately 1.5; additions of large Ba and Pb ions (as BaO and PbO) to a glass increase n significantly, e.g., highly leaded glasses containing 90 wt.% PbO have an index of refraction of approximately 2.1

Optical Properties of Nonmetals

- For crystalline ceramics with cubic crystal structures and for glasses, the index of refraction is independent of crystallographic direction (i.e., it is isotropic)
- Noncubic crystals, however, have an anisotropic n — i.e. the refractive index is greatest along the directions that have the highest density of ions
- Higher the index of refraction of the solid, the greater is the reflectivity; for typical silicate glasses, the reflectivity is approximately 0.05
- Just as the index of refraction of a solid depends on the wavelength of the incident light, so does the reflectivity vary with wavelength
- Reflection losses for lenses and other optical instruments are minimized significantly by coating the reflecting surface with very thin layers of dielectric materials such as magnesium fluoride (MgF_2)
- Nonmetallic materials may be opaque or transparent to visible light; if transparent, they often appear colored
- In principle, light radiation is absorbed in this group of materials by two basic mechanisms that also influence the transmission characteristics of these nonmetals

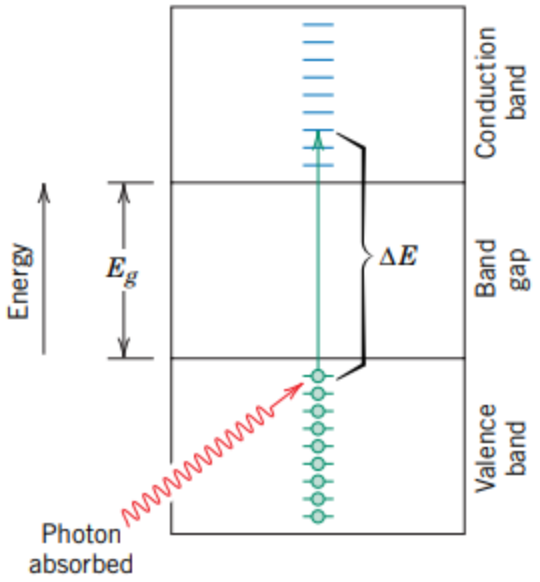
Optical Properties of Nonmetals

- One of these mechanisms is **electronic polarization**; absorption by electronic polarization is important only at light frequencies in the vicinity of the relaxation frequency of the constituent atoms (the frequency at which the dielectric loss factor (∈”) reaches a maximum, for a dielectric material that has no static (d.c.) conductivity and that is subjected to an alternating electromagnetic field)

- The other mechanism involves **valence band–conduction band electron transitions**, which depend on the electron energy band structure of the material

- Absorption of a photon of light may occur by the promotion or excitation of an electron from the nearly filled valence band, across the band gap, and into an empty state within the conduction band

Fig. 5 Photon absorption for nonmetallic materials in which an electron is excited across the band gap, leaving behind a hole in the valence band



Optical Properties of Nonmetals

- These excitations with the accompanying absorption can take place only if the photon energy is greater than that of the band gap E_g, i.e. if

$$h\nu > E_g$$

or in terms of wavelength

$$\frac{hc}{\lambda} > E_g$$

The minimum wavelength for visible light, $\lambda(\text{min})$, is about 0.4 μm, and because $c = 3 \times 10^8$ m/s and $h = 4.13 \times 10^{-15}$ eV · s, the maximum band gap energy $E_g(\text{max})$ for which absorption of visible light is possible is

$$E_g(\text{max}) = \frac{hc}{\lambda(\text{min})}$$

$$= \frac{(4.13 \times 10^{-15}\ \text{eV} \cdot \text{s})(3 \times 10^8\ \text{m/s})}{4 \times 10^{-7}\ \text{m}}$$

$$= 3.1\ \text{eV}$$

- In other words, no visible light is absorbed by nonmetallic materials having band gap energies greater than about 3.1 eV; these materials, if of high purity, appear **transparent and colorless**

Optical Properties of Nonmetals

However, the maximum wavelength for visible light, $\lambda(\text{max})$, is about 0.7 μm; computation of the minimum band gap energy $E_g(\text{min})$ for which there is absorption of visible light gives

$$E_g(\text{min}) = \frac{hc}{\lambda(\text{max})}$$

$$= \frac{(4.13 \times 10^{-15}\ \text{eV} \cdot \text{s})(3 \times 10^{8}\ \text{m/s})}{7 \times 10^{-7}\ \text{m}}$$

$$= 1.8\ \text{eV}$$

- This result means that all visible light is absorbed by valence band–to–conduction band electron transitions for semiconducting materials that have band gap energies less than about 1.8 eV; thus, these materials are **opaque**

- Only a portion of the visible spectrum is absorbed by materials having band gap energies between 1.8 and 3.1 eV; consequently, these materials appear **colored**

- Every nonmetallic material becomes opaque at some wavelength, which depends on the magnitude of its E_g; for example, diamond, having a band gap of 5.6 eV, is opaque to radiation having wavelengths less than about 0.22 μm

Optical Properties of Nonmetals



Fig. 6 The transmission of light through a transparent medium for which there is reflection at the front and back faces, as well as absorption within the medium

- For an incident beam of intensity I_0 that impinges on the front surface of a specimen of thickness l and absorption coefficient β, the transmitted intensity at the back face I_T is

$$I_T = I_0(1 - R)^2e^{-\beta l}$$

where R is the reflectance; for this expression, it is assumed that the same medium exists outside both front and back faces

- Thus, the fraction of incident light that is transmitted through a transparent material depends on the losses that are incurred by absorption and reflection (also, the respective fractions depend on light wavelength, e.g. 0.90T+0.05A+0.05R at 0.4 μm, while at 0.55 μm, 0.50T+0.48A+0.02R)